

TASVIRLARNI TAHLIL QILISH VA ANGLASH · Amaliy mashg'ulot (Practice)

12-amaliy mashg'ulot. Stereo ko'rish va chuqurlik

2026–2027 o'quv yili

Mavzu. Rektifikatsiyalangan stereo, disparity, $Z = fB/d$ va uning xatosi; epipolar geometry (essential va fundamental matrix), rektifikatsiya, triangulyatsiya; moslik narxлари (SAD, NCC, census), cost volume, WTA, subpiksel, left–right consistency, SGM; StereoBM, StereoSGBM, `reprojectImageTo3D`.

Maqsadlar. Talaba: chuqurlik va uning xatosini qo'lda hisoblab stereo tizimni loyihalaydi; berilgan F bilan epipolar shartni tekshiradi; block matching, parabola subpiksel, LR check va SGM ni NumPy da yozadi; OpenCV StereoBM/StereoSGBM ni sun'iy va real (Aloe, Middlebury) juftlikda ground truth bo'yicha baholaydi.

Muhit. Python 3.11 yoki 3.12, faqat CPU. `python -m venv .venv`, so'ng `pip install -r requirements.txt` (numpy 2.4.4, opencv-python-headless 4.13.0.92). Kod boshida `np.random.seed(42)`, `cv2.setRNGSeed(42)`, `cv2.setNumThreads(1)`; tasodifiylik — `np.random.default_rng(42)`. Real juftlik `urllib.request` bilan yuklanadi va `./data/` da keshlanadi. Sonlar 4 xonagacha.

Baholash (100 ball). A (1–13): har biri 2 ball; B (14–17): har biri 4 ball; C (18–30): 25, 26 va 27-topshiriq 6 balldan, qolganlari 4 balldan. Kod — bitta `solutions.py`, har topshiriq alohida funksiya.

A. Qo'lda hisoblash

- Chuqurlik.** $f = 800$ px, $B = 0,15$ m, $c_x = 400$, $c_y = 300$. $(x_L, y) = (560; 330)$, $d = 40$ px: Z, X, Y ? $d = 41$ bo'lsa Z qanchaga o'zgaradi?
- Z^2 qonuni.** Shu tizimda $\Delta d = 0,5$ px. $Z = 4$ va $Z = 20$ m uchun aniq $|\Delta Z| = Z^2 \Delta d / (fB + Z \Delta d)$ va yaqinlashuv $Z^2 \Delta d / (fB)$; nisbiy xato.
- Loyihalash.** $Z = 15$ m da $\Delta Z \leq 0,2$ m, subpiksel $\Delta d = 0,25$ px. Minimal fB ? $f = 1250$ px bo'lsa B ? `numDisparities=128` da eng yaqin o'lchanadigan masofa?
- Noaniqlik oralig'i.** $fB = 120$ px·m, $d = 3$ px. Z va $d \pm 1$ px ga mos oraliq. Nega nosimmetrik?
- Epipolar shart.** $F = \begin{pmatrix} 0 & -0,001 & 0,1 \\ 0,001 & 0 & -0,3 \\ -0,1 & 0,3 & 0 \end{pmatrix}$, $\mathbf{x}_L = (100; 200; 1)$. (a) $\ell_R = F\mathbf{x}_L$; $(200; 150)$ shu chiziqdami? $(200; 155)$ ning chiziqqacha masofasi? (b) Epipol ($F\mathbf{e} = 0$) va rank F . (c) Rektifikatsiyalangan $F = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$ uchun $\mathbf{x}_R^T F \mathbf{x}_L = 0$ nimani bildiradi?
- Essential matrix.** $R = I$, $\mathbf{t} = (-0,1; 0; 0)$ m. $E = [\mathbf{t}]_{\times} R$, singular qiymatlari. Normallangan $\hat{\mathbf{x}}_L = (0,2; 0,1; 1)$, $\hat{\mathbf{x}}_R = (0,15; 0,1; 1)$ uchun $\hat{\mathbf{x}}_R^T E \hat{\mathbf{x}}_L$ va Z .
- SAD va NCC.** Chap oyna $(12, 40, 25)$; o'ngdagi nomzodlar $d = 0$: $(30, 18, 44)$, $d = 1$: $(14, 37, 27)$, $d = 2$: $(40, 26, 10)$, $d = 3$: $(41, 24, 9)$. SAD va WTA; $d = 1$ va $d = 2$ uchun NCC.
- Census.** Chap $\begin{pmatrix} 12 & 30 & 8 \\ 25 & 20 & 40 \\ 18 & 22 & 5 \end{pmatrix}$, o'ng $\begin{pmatrix} 15 & 28 & 9 \\ 19 & 21 & 45 \\ 25 & 23 & 4 \end{pmatrix}$. Bitlar (qator bo'yicha, markazsiz, qo'shni $<$ markaz $\Rightarrow$ 1), Hamming masofasi; chap oynaga $+20$ qo'shilsa?
- Subpiksel.** $d_0 = 17$: (a) $c = (9, 3, 5)$; (b) $c = (4, 2, 10)$. $\delta = \frac{c_- - c_+}{2(c_- - 2c_0 + c_+)}$ va d^* .
- LR check.** $D_L(60..64) = (15, 15, 15, 22, 22)$; $D_R(45..47) = (15; 15; 14, 6)$, $D_R(41) = 22$, $D_R(42) = 16$. $\tau = 1$ da qaysi piksellar ishonchli?
- SGM qadami.** Oldingi $L_r = (4, 1, 5, 8)$, joriy $C = (2, 3, 1, 6)$, $P_1 = 1$, $P_2 = 3$. Yangi $L_r(\cdot)$ va tanlangan d .
- OpenCV amaliyoti va soft-argmin.** StereoSGBM `blockSize=5`, kulrang: tavsiya P_1, P_2 ; chiqishda 412 — disparity? `numDisparities=100` ni qanday tuzatish kerak? Soft-argmin: $c = (3, 1, 2, 4)$ uchun $\sigma(-c)$ va $\hat{d}$.
- Ko'p bazali stereo.** $f = 600$ px, $Z = 8$ m, $\Delta d = 0,25$; $B_1 = 0,1$ m, $B_2 = 0,3$ m. σ_1, σ_2 , teskari dispersiya bilan birlashgan σ ; $\Delta x = 0,5$ px da lateral xato ΔX .

B. Keltirib chiqarish va tushunchalar

- O'xshash uchburchaklardan $Z = fB/d$, X, Y ni chiqaring; aniq $|\Delta Z|$ va Z^2 qonunini isbotlang; nisbiy xato $\Delta Z/Z \approx Z \Delta d / (fB)$.
- $\mathbf{X}_R = R\mathbf{X}_L + \mathbf{t}$ dan epipolar shart $\hat{\mathbf{x}}_R^T E \hat{\mathbf{x}}_L = 0$, $E = [\mathbf{t}]_{\times} R$ va $F = K_R^{-T} E K_L^{-1}$ ni chiqaring. E ning singular qiymatlari ($\|\mathbf{t}\|, \|\mathbf{t}\|, 0$) ekanini ko'rsating.
- Parabola subpiksel formulasini va $|\delta| \leq \frac{1}{2}$ ni isbotlang. NCC ning $I \mapsto aI + b$ ($a > 0$) ga invariantligini isbotlang; census ning monoton o'zgarishlarga invariantligi.

17. Qisqa javob: SGM energiyasi, $P_1 < P_2$ roli; nega to'qimasiz sohada WTA adashadi; occlusion va LR check; nega MAE o'rniga bad-pixel rate; qamrov+xato juftligi; `reprojectImageTo3D` da Q_{43} ishorasi.

C. Dasturlash

Sun'iy juftlik (21–26, 28-topshiriqlar; aynan shu generator, haqiqiy disparity D):

```
H, W, DMAX = 120, 200, 48
V = np.s_[:, DMAX + 5:W - 5] # baholash sohasi
TL = np.s_[:, 92:118] # to'qimasiz soha (chap tasvirda)
def make_pair(seed=42):
    rng = np.random.default_rng(seed)
    D = np.full((H, W), 10.4); D[30:90, 60:130] = 20.7; D[45:75, 140:175] = 39.3
    tex = cv2.GaussianBlur(rng.uniform(0, 255, (H, W + 80)), (0, 0), 1.2)
    tex[:, 130:160] = 128 # to'qimasiz soha
    xs = np.arange(W, dtype=np.float64)
    L = np.zeros((H, W), np.float32); R = np.zeros((H, W), np.float32)
    for y in range(H): # R(x - d) = L(x)
        L[y] = np.interp(xs + 40, np.arange(W + 80), tex[y])
        R[y] = np.interp(xs + 40 + D[y], np.arange(W + 80), tex[y])
    return L, R, D
```

Umumiy: cost volume C shakli ($D_{\max} + 1, H, W$), yetishmaydigan ustunlar 10^9 ; har d uchun `cv2.boxFilter(..., normalize=False)`. WTA + parabola: d_0 ni $[1, D_{\max} - 1]$ ga qirgib, δ ni $[-\frac{1}{2}, \frac{1}{2}]$ ga; maxraj ≤ 0 bo'lsa $\delta = 0$. bad-1 — $|d - D| \geq 1$ ulushi (V sohada). OpenCV natijasi /16, yaroqli — $d > 0$.

18. `depth(d,f,B)`, `depth_error(Z,fB,dd)` → (aniq, yaqin). A1–A4 ni tekshiring; $fB = 120$, $\Delta d = 0,5$, $Z \in \{1, 2, 4, 8, 20\}$ jadvali: d , aniq, yaqin, nisbiy xato (%).
19. **Epipolar geometriya.** `skew(t)`, `fundamental(K1,K2,R,t)`, `epi_dist(F,xL,xR)` (px), `DLT triangulate(P1,P2,x1,x2)`. A5 ni tekshiring. Sahna: $K = [[500, 0, 320], [0, 500, 240], [0, 0, 1]]$, $R = \text{Rodrigues}((0, 0.05, 0.02))$, $t = (-0.12, 0.01, 0)$; 50 nuqta: `rng=default_rng(42)`, $X, Y, Z = \text{uniform}(-1, 1)$, `uniform(-0.7, 0.7)`, `uniform(2, 6)` (har biri 50 ta, shu tartibda). E singular qiymatlari; haqiqiy F da o'rtacha masofa; so'ng xuddi shu `rng` bilan $\mathcal{N}(0, 0.5^2)$ shovqin (avval chap, keyin o'ng) va `findFundamentalMat(FM_8POINT)`: shovqinli va toza nuqtalarda o'rtacha masofa; DLT max xatosi.
20. **Rektifikatsiya.** 19-sahna uchun `stereoRectify(K, 0, K, 0, (640, 480), R, t, alpha=0)`: T_x, f', Q_{32}, Q_{23} . Nuqtalarni `undistortPoints(..., R=R1/R2, P=P1/P2)` bilan o'tkazing: $|y_L - y_R|$ oldin/keyin; $Z = Q_{23}/(Q_{32}d)$ ni rektifikatsiyalangan chap kameradagi haqiqiy chuqurlik bilan solishtiring.
21. **SAD block matching.** `sad_volume(A,B,w)` va `wta_subpix(C)` → (butun, parabola). $w \in \{5, 9, 15, 21\}$: bad-1, to'qimasiz sohada bad-1, median va MAE (parabola bilan).
22. **Subpiksel.** $w = 9$: butun va parabola uchun median, bad-1, to'g'ri piksellarda ($|e| < 1$) MAE. A9 ni tekshiring.
23. **LR check.** O'ngdan chapga: `wta_subpix(sad_volume(R[:,::-1], L[:,::-1], 9))` natijasini qayta akslantirib; $|D_L(x) - D_R(x - \text{round}(D_L))| \leq 1$. Ishonchli ulush, MAE (hammasi/ishonchli), ishonchlilarda bad-1. A10 ni tekshiring.
24. **Yoritishga chidamlilik.** `census(I,r=2)` (24 bit, `np.pad(mode='edge')`), `census_volume` (Hamming, `np.unpackbits`), `ncc_volume` ($1 - \text{NCC}$, `cv2.blur w \times w`). $w = 9$ da chap tasvir (a) asl, (b) $L + 20$, (c) $1, 2L + 10$: SAD, census, NCC bad-1. A7, A8 ni tekshiring.
25. **SGM.** `sgm_path(C,P1,P2)` — chapdan o'ngga 1D dinamik dasturlash (ma'ruza formulasi), A11 ni tekshiring. $C = \text{sad_volume}(L, R, 5)/25$ (10^9 o'rniga 255), $P_1 = 4$, $P_2 = 32$, ikki yo'nalish (chap→o'ng va o'ng→chap) yig'indisi. WTA bilan bad-1 va to'qimasiz bad-1 ni solishtiring.
26. **OpenCV.** `StereoBM.create(64, 9)` va `StereoSGBM.create(0, 64, 5, P1=200, P2=800, uniquenessRatio=10, speckleWindowSize=100, speckleRange=2)` (uint8 ga clip). Qamrov, MAE, bad-1 (yaroqlilarda), minimal qiymat.
27. **Real juftlik (Aloe).** <https://raw.githubusercontent.com/opencv/opencv/4.x/samples/data/aloeL.jpg>, `aloeR.jpg`, `aloeGT.png` (ground truth disparity, 0 — noma'lum). Kulrang, `INTER_AREA` bilan 2 marta kichraytirib (GT — `INTER_NEAREST` va /2). `StereoBM(128, 15)` va 26-dagi SGBM (`numDisparities=128`): ma'lum piksellarga nisbatan qamrov, median xato, bad-1, bad-2. SGBM ga LR check (akslantirilgan juftlik): qamrov va bad-2.
28. **3D ga o'tish.** $Q = [[1, 0, 0, -W/2], [0, 1, 0, -H/2], [0, 0, 0, f], [0, 0, \pm 1/B, 0]]$, $f = 500$, $B = 0.12$; `reprojectImageTo3D(D, Q)`: $y = 60$, $x \in \{30, 100, 150\}$ dagi Z ikki ishora uchun; fB/d va $X(60, 150)$. A6 ni tekshiring.
29. **Monte Carlo.** $fB = 60$, $\sigma_d = 0.25$, $Z \in \{2, 5, 10, 20\}$: 10^5 ta $d = fB/Z + \mathcal{N}(0, \sigma_d^2)$ (bitta `rng`), $\hat{Z} = fB/d$ ning std si, $Z^2\sigma_d/(fB)$ va siljish (bias). A13 ni tekshiring.

30. Soft-argmin. $\text{soft_argmin}(c) \rightarrow (\sigma(-c), \hat{d})$. A12 ni tekshiring; ikki choʻqqili $c = (1, 6, 6, 1, 2)$ uchun $\hat{d}$ va argmin ni solishtiring; SGBM parametrlari hisobi.

Hujjatlar.

- https://docs.opencv.org/4.x/d9/d0c/group__calib3d.html
- https://docs.opencv.org/4.x/d2/d85/classcv_1_1StereoSGBM.html
- https://docs.opencv.org/4.x/d9/dba/classcv_1_1StereoBM.html
- https://docs.opencv.org/4.x/dd/d53/tutorial_py_depthmap.html
- https://docs.opencv.org/4.x/da/de9/tutorial_py_epipolar_geometry.html
- <https://docs.python.org/3/library/urllib.request.html>
- <https://vision.middlebury.edu/stereo/data/scenes2006/>